

Challenge 4: Sidorenko Conjecture for Bipartite Graphs with Bounded Sides

Definition 1 (Graph homomorphism density). Let G and H be simple graphs.

1. A function $f : V(H) \rightarrow V(G)$ is a *graph homomorphism* from H to G if for each edge $uv \in E(H)$, we have $f(u)f(v) \in E(G)$.
2. Let $\text{hom}(H, G)$ be the number of graph homomorphisms $H \rightarrow G$.
3. The *homomorphism density* of H in G is the fraction

$$t_H(G) := \frac{\text{hom}(H, G)}{|V(G)|^{|V(H)|}}$$

Conjecture 1.1 (Sidorenko's Conjecture [1]). For every bipartite graph H and for any graph G , we have:

$$t_H(G) \geq t_{K_2}(G)^{|E(H)|}$$

We are interested in the special case of the conjecture where H is a bipartite graph in which each side has at most r vertices.

Open Problem 1.2 (Scaled Sidorenko's Conjecture). Let $r \in \mathbb{N}$. For every bipartite graph H that admits a bipartition $V(H) = X \cup Y$ with $|X| \leq r$ and $|Y| \leq r$ and for any graph G , we have:

$$t_H(G) \geq t_{K_2}(G)^{|E(H)|}$$

References

- [1] A. F. Sidorenko. "A Correlation Inequality for Bipartite Graphs". In: *Graphs and Combinatorics* 9.2–4 (1993), pp. 201–204. DOI: 10.1007/BF02988307. URL: <https://doi.org/10.1007/BF02988307>.